

PORT ELIZABETH INTL, South Africa

WMO: 688420

Lat: 33.9864S

Lon: 25.6164E

Elev: 207

StdP: 14.59

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	41.4	44.2	30.5	24.8	61.4	33.8	28.6	57.9	35.3	58.4	31.5	57.6	4.3	280	0.529

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	13.6	85.1	65.8	81.6	67.0	79.1	67.5	72.8	78.5	71.6	77.1	70.4	75.8	10.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.3	116.3	75.5	69.8	110.4	74.5	68.3	104.8	73.2	36.6	78.7	35.5	77.5	34.5	75.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 32.4	28.8	25.7	DB	35.5	97.1	2.2	3.3	33.9	99.5	32.6	101.4	31.3	103.3	29.7	105.7	(4)
(5)			WB	34.0	75.2	2.5	1.4	32.3	76.2	30.8	77.0	29.4	77.8	27.6	78.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.9	70.5	71.0	68.8	65.1	61.8	58.1	57.3	58.3	59.9	62.7	65.1	68.1	(6)	
	DBStd	6.17	3.40	3.41	3.67	4.66	4.32	4.32	4.42	4.25	3.85	4.16	3.82	3.56	(7)	
	HDD50	1	0	0	0	0	0	0	1	0	0	0	0	0	(8)	
	HDD65	1163	3	2	9	54	118	211	242	213	159	96	44	12	(9)	
	CDD50	5059	635	588	582	452	367	245	229	258	298	393	452	560	(10)	
	CDD65	742	172	169	126	56	20	5	5	6	7	23	46	107	(11)	
	CDH74	2804	544	573	372	231	209	115	129	114	51	94	100	272	(12)	
	CDH80	514	44	67	61	81	75	22	27	40	18	32	20	27	(13)	
(14) Wind	WSAvg	11.8	12.7	12.4	11.8	10.8	10.0	10.1	10.2	11.4	12.5	13.3	13.4	12.7	(14)	
(15) Precipitation	PrecAvg	23.6	1.3	1.4	2.0	2.0	2.1	2.2	1.9	2.6	2.0	2.2	2.1	1.5	(15)	
	PrecMax	41.7	4.7	4.4	12.2	5.8	6.5	8.3	8.5	10.0	8.0	8.4	9.4	7.2	(16)	
	PrecMin	15.7	0.2	0.2	0.2	0.1	0.1	0.1	0.0	0.2	0.4	0.4	0.3	0.2	(17)	
	PrecStd	5.9	1.1	0.9	1.9	1.4	1.8	1.9	1.8	2.1	1.4	1.5	1.6	1.3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.1	85.8	86.3	89.8	88.0	82.8	83.9	85.7	80.6	82.8	81.0	83.7	(19)	
		MCWB	71.8	72.5	69.8	67.0	62.3	57.7	58.7	58.7	61.5	65.1	66.4	70.3	(20)	
	2%	DB	80.7	81.7	80.4	79.9	80.3	77.1	77.4	75.3	72.0	75.0	76.2	78.9	(21)	
		MCWB	70.9	71.2	69.2	65.6	61.2	57.3	56.5	57.4	61.6	64.2	66.4	68.9	(22)	
	5%	DB	78.8	79.1	77.4	75.0	73.3	71.7	71.2	68.9	69.5	71.6	73.7	76.9	(23)	
		MCWB	69.5	69.8	67.9	65.5	60.5	56.1	55.4	58.1	60.9	63.3	64.8	67.8	(24)	
	10%	DB	77.1	77.3	75.4	71.9	69.7	67.4	66.3	66.2	67.2	69.6	71.8	75.1	(25)	
		MCWB	68.5	68.8	66.9	63.9	60.6	56.5	56.0	58.0	59.7	62.1	63.9	66.6	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	74.4	74.6	72.7	70.1	66.9	63.1	62.1	64.0	65.6	68.0	69.9	72.9	(27)	
		MCDB	80.4	81.2	79.9	80.4	73.7	69.4	70.1	71.1	71.7	76.7	76.2	77.9	(28)	
	2%	WB	72.6	72.9	70.9	68.3	64.5	60.9	60.3	62.1	63.9	65.8	68.1	70.9	(29)	
		MCDB	77.8	78.7	77.3	75.4	71.5	67.8	67.7	67.1	69.2	72.0	73.3	76.1	(30)	
	5%	WB	71.2	71.7	69.5	66.9	63.2	59.4	59.1	60.7	62.5	64.6	66.8	69.3	(31)	
		MCDB	76.4	77.1	75.1	72.8	69.7	66.5	65.8	65.8	67.4	69.9	71.8	74.6	(32)	
	10%	WB	70.0	70.5	68.3	65.5	61.9	58.2	57.7	59.4	61.0	63.4	65.3	68.1	(33)	
		MCDB	75.2	75.4	73.5	70.7	68.0	65.2	64.2	64.8	65.7	68.1	70.2	72.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.4	13.6	14.1	15.9	18.1	20.5	20.1	18.0	15.7	14.2	13.9	13.6	(35)	
		MCDBR	15.1	16.1	17.6	21.4	27.0	30.2	31.3	25.6	20.2	17.6	16.6	16.1	(36)	
	5% WB	MCWBR	7.6	7.8	8.4	10.0	12.9	14.6	14.8	13.3	11.4	9.5	8.9	8.0	(37)	
		MCWBR	13.3	14.5	15.2	18.0	20.0	22.0	21.9	19.5	17.3	15.9	15.0	14.5	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.389	0.392	0.377	0.353	0.339	0.327	0.329	0.367	0.385	0.399	0.380	0.384	(40)	
		taud	2.470	2.471	2.496	2.536	2.548	2.555	2.547	2.431	2.373	2.368	2.457	2.472	(41)	
	Ebn at Noon		301	293	286	275	262	257	263	266	278	288	302	304	(42)	
		Edh at Noon	37	36	33	28	24	23	24	30	36	39	37	37	(43)	
(44) All-Sky Solar Radiation	RadAvg		2205	1972	1649	1243	946	827	886	1138	1510	1819	2099	2280	(44)	
	RadStd		114	91	108	63	45	48	57	69	90	132	142	142	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		+0.68	+1.24	+1.51	N/A	N/A	N/A	N/A	-153	+256	+81	(46)
(47) Regional (5 neighbors)		+0.68	+1.24	+1.51	N/A	N/A	N/A	N/A	-153	+256	+81	(47)

Nomenclature: See separate page